FAST, National University of Computing and Emerging Sciences, Islamabad, Pakistan

ICD Integrated Circuit Design Lab

Energy Efficient Accelerator Design for AI on Edge Applications

Team Members

- Ahmad Shaban
- Kousar Gul
- Ume Kalsoom
- Awab Younas
- Muhammad Mirza
- Dr. Khurram Javed
- Eeman
- Osama Mirza
- Dr. Hassan Saif
- Fajar Waseem
- Salman Qazi
- Dr. Rashid Ramzan
- Humail Nawaz
- Sana Nazir

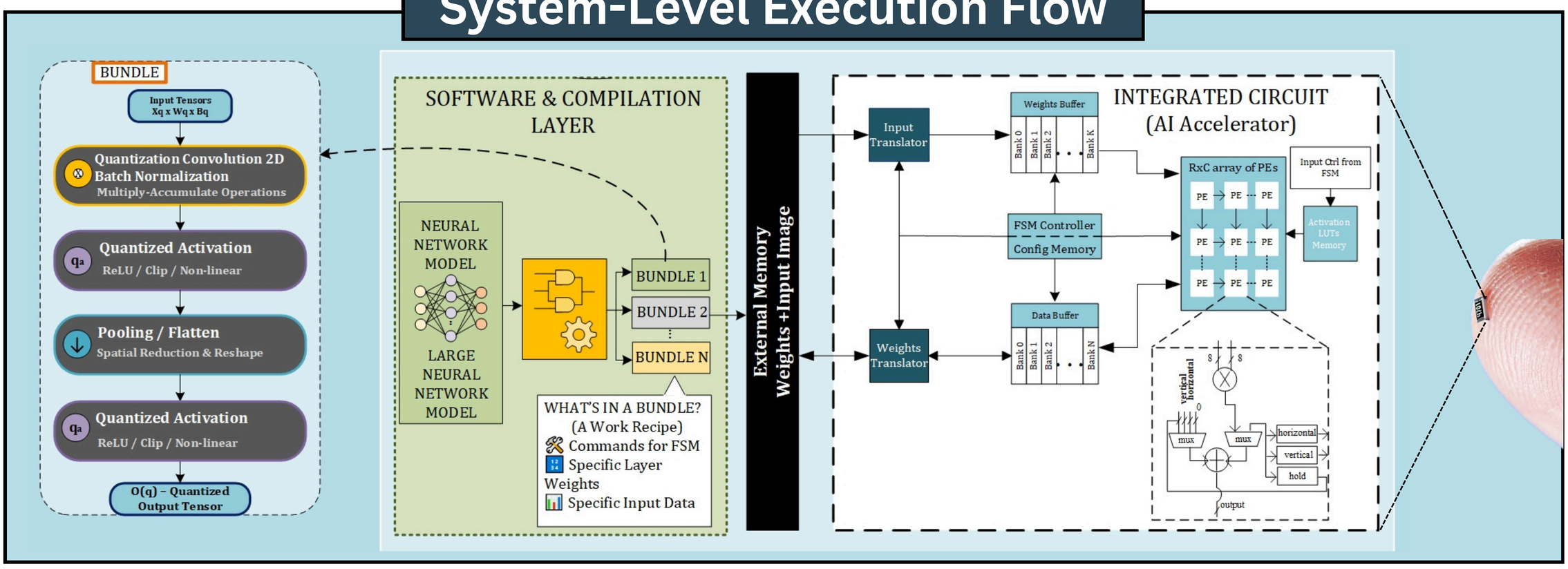
Project Objectives

- Efficient Reconfigurable CNN Accelerator
- Optimizing PE using a Modified Booth multiplier and compression reduction tree for parallel MAC operations
- A parameterizable R×C PE array supporting sub-8-bit quantized convolution execution utilizing maximum data reuse strategy
- Reconfigurable CNN Accelerator deployment on FPGA/ASIC

Abstract

This work presents an energy-efficient CNN accelerator that optimizes the Processing Element using a Modified Booth multiplier and reduction tree to reduce delay, power, and area. An FSM controller manages a reconfigurable R×CR times CR×C PE array to perform parallel MAC operations, while a LUT applies quantized activation functions to the outputs.

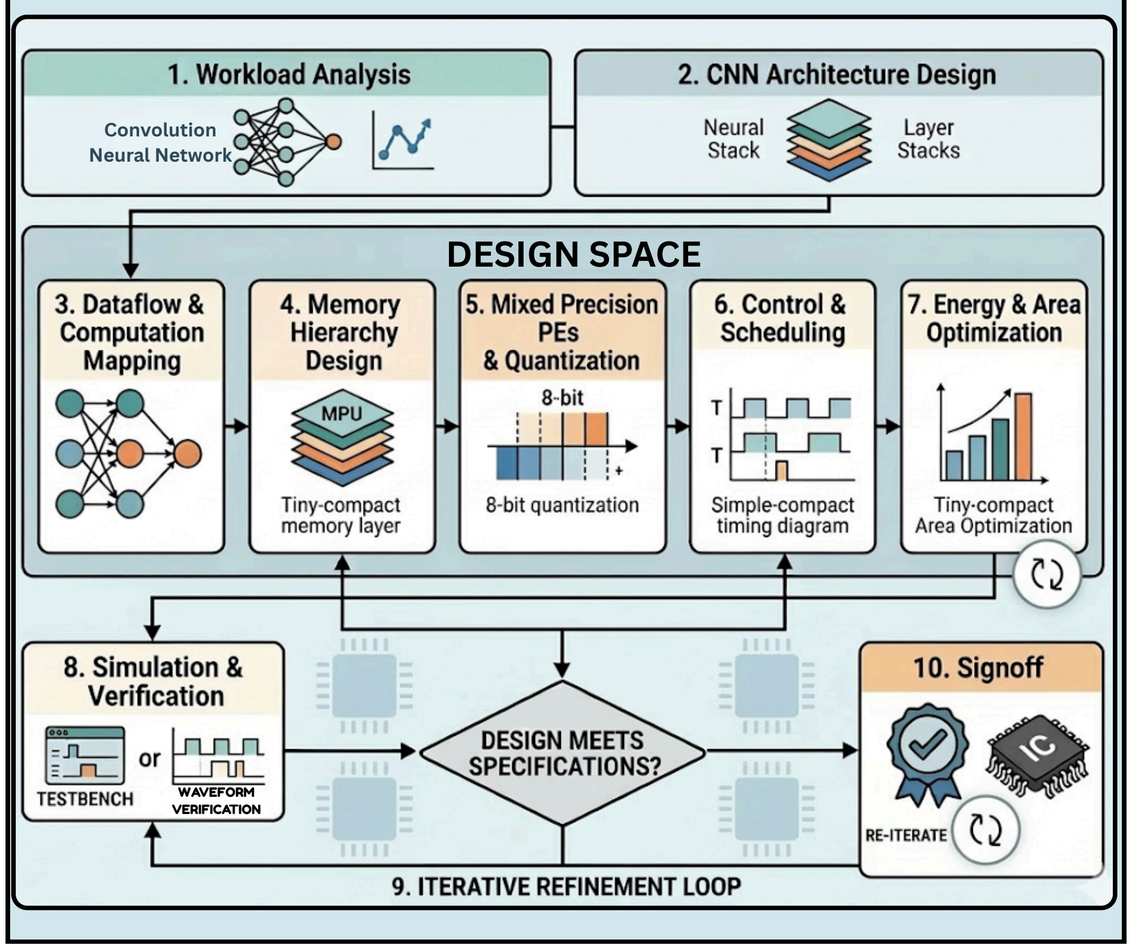
System-Level Execution Flow



Results

Multiplier Architecture	Power (µW)	Area (µm²)
WTM (AND + CSA + CPA)	50.2	74317.56
MBE-R2 Multiplier	22.2	67719.6
MBE-R4 + GPC	47.93	61121.64
MBE-R4 Multiplier	32.8	54523.68
Approximate Multiplier	47.9	47925.72
MBE-R8 + Compressor + CLA	419	41327.76
MBE-R8 + GPC	233.25	34729.8
MBE-R8 Multiplier	196.7	28131.84
MBE-R4 (PP Generation)	11.4	21533.88
MBE-R8 (PP Generation)	9.47	14935.92
MBE-R4 + Compressor + CPA	264	8337.96
WTM (Basic Adders: AND + CSA)	65	1740

Methodology



Future Work

- Integrate the optimised PE into a full R×C CGRA fabric and evaluate end-to-end CNN inference (ResNet-50, VGG) on FPGA with measured throughput and energy.
- Target ASIC tapeout on TSMC 65 nm and benchmark power, performance and area against commercial accelerators.